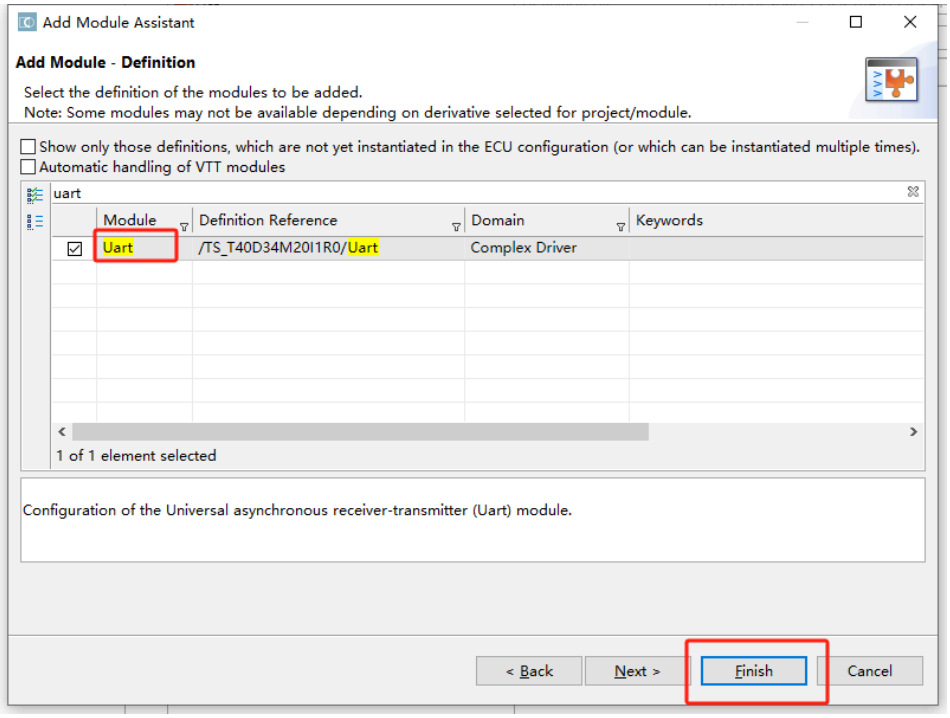
9、添加UART模块_使能所有外设时钟

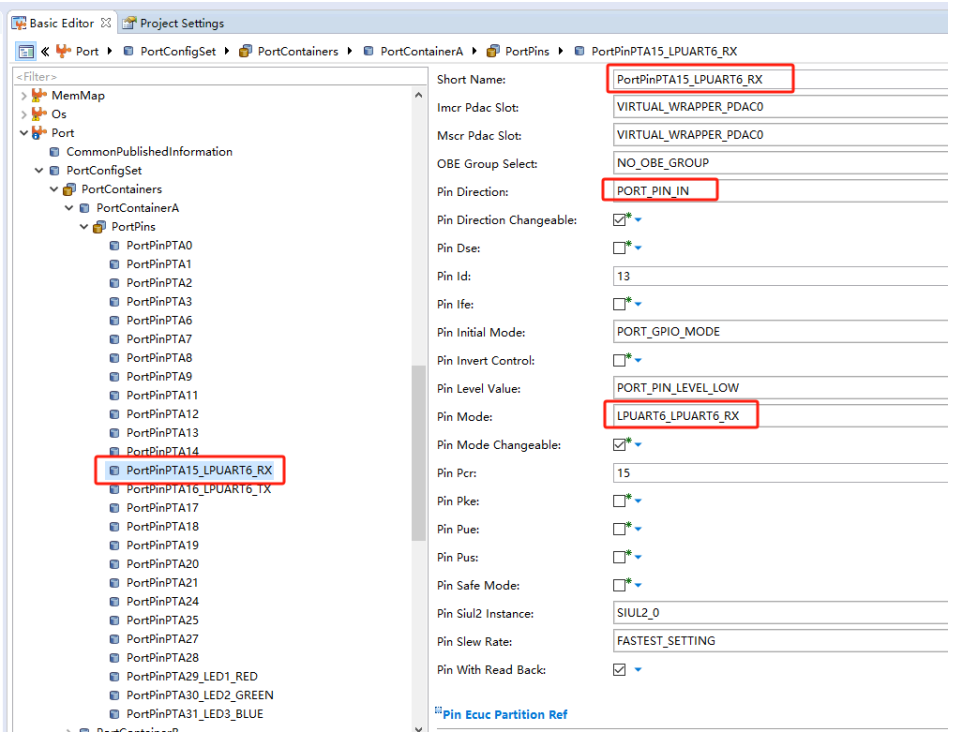
卖家'漩涡遨游aa' 专属资料，非在漩涡遨游aa 购买资料，不提供答疑(一次购买仅提供有限次数10次答疑)

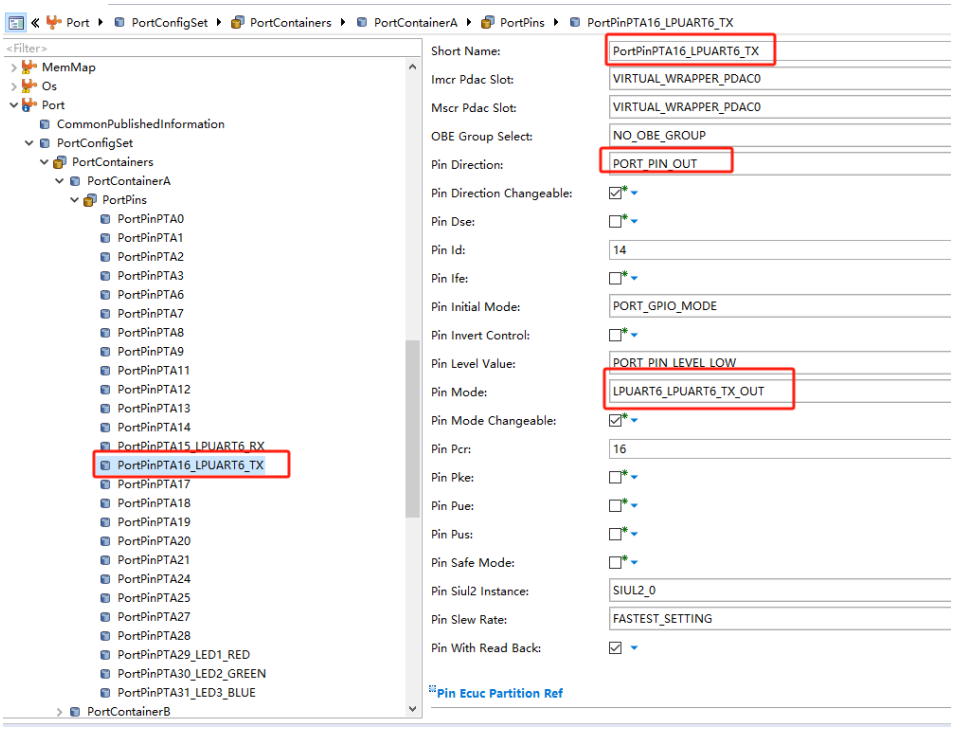
一、添加uart模块



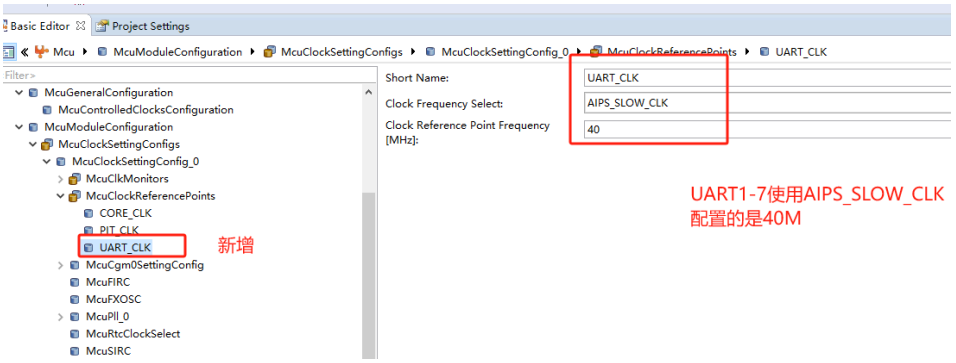
二、修改Port

修改UART6，PTA15/16

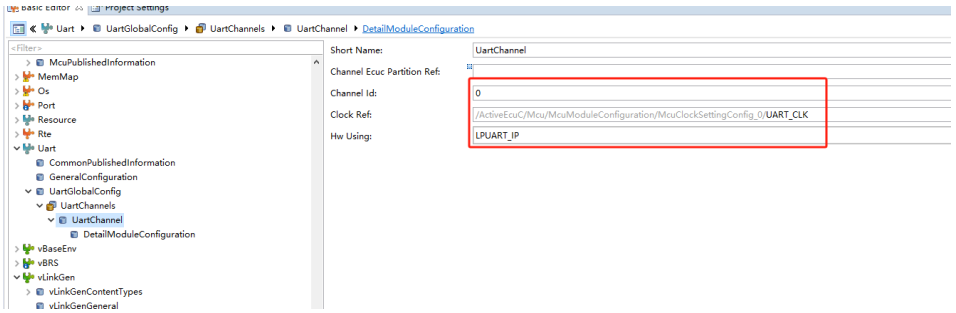


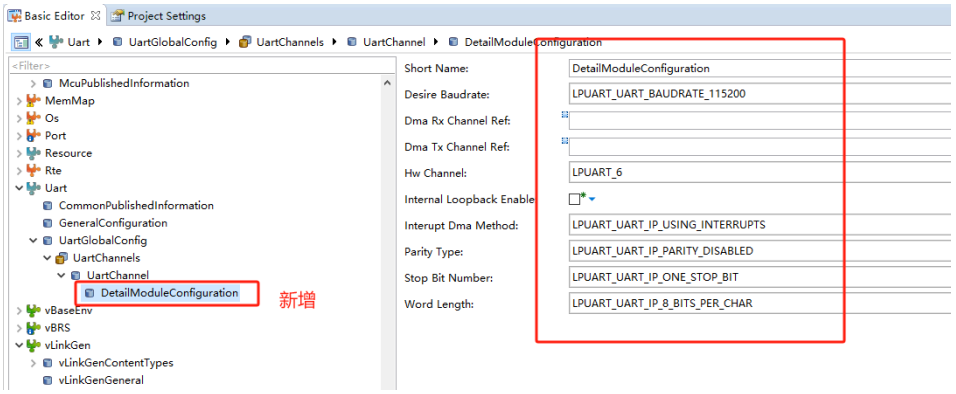


三、mcu添加UART_CLK参考时钟

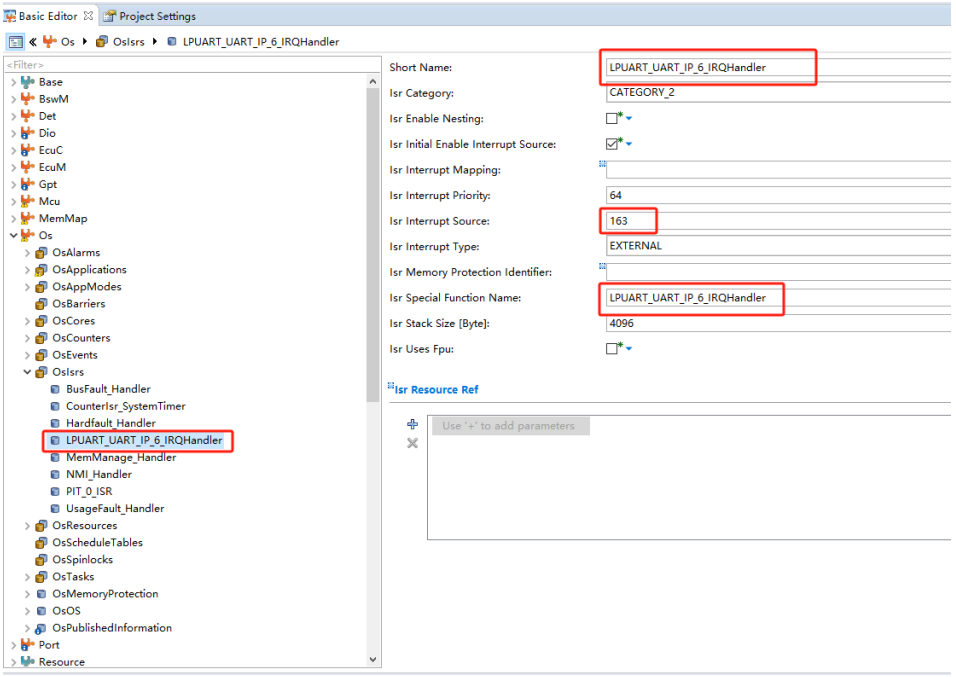


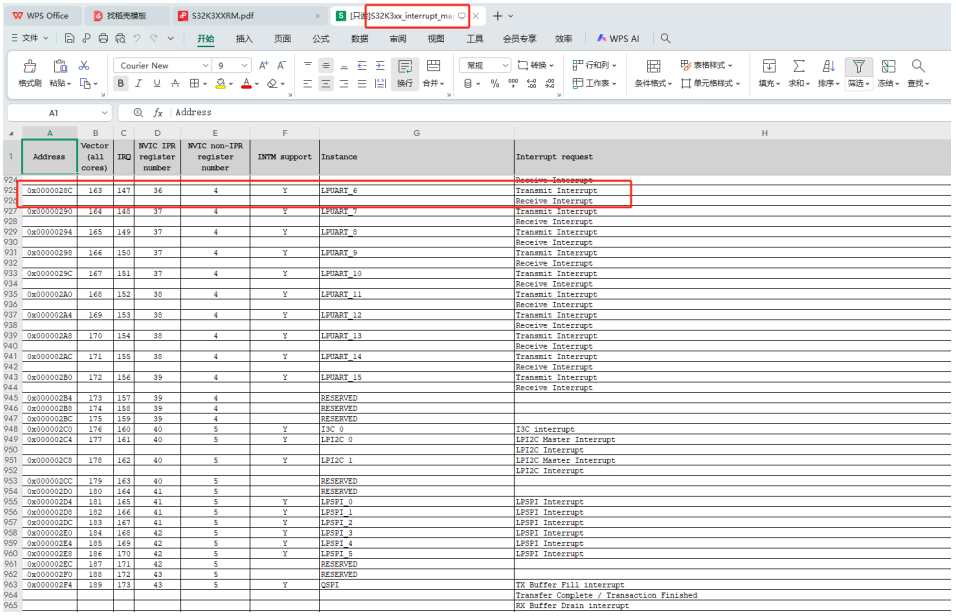
四、修改uart模块





五、添加中断

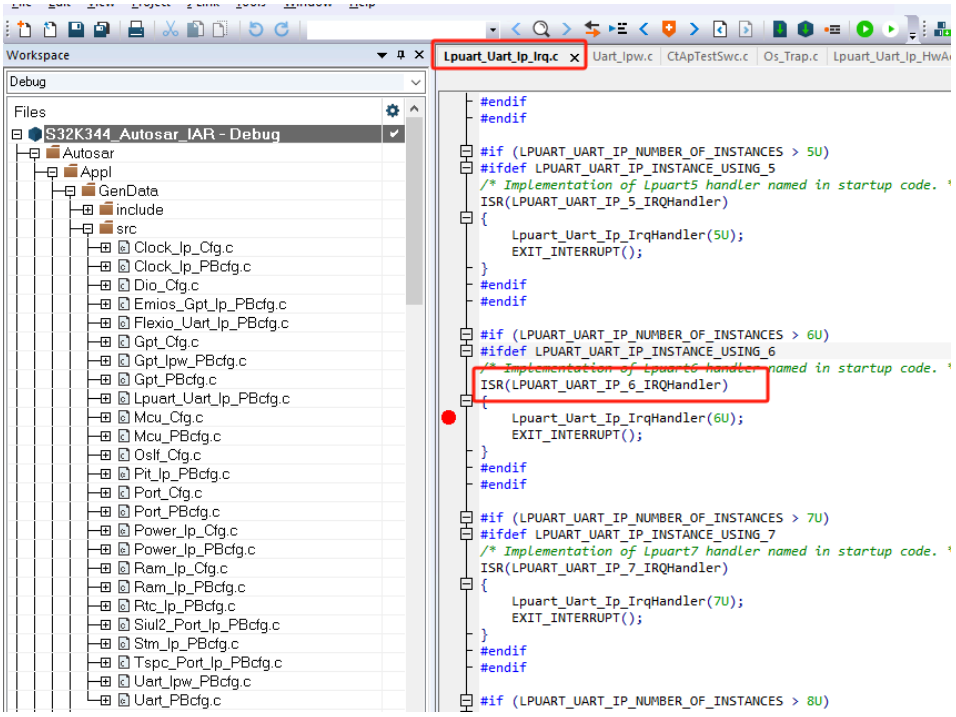




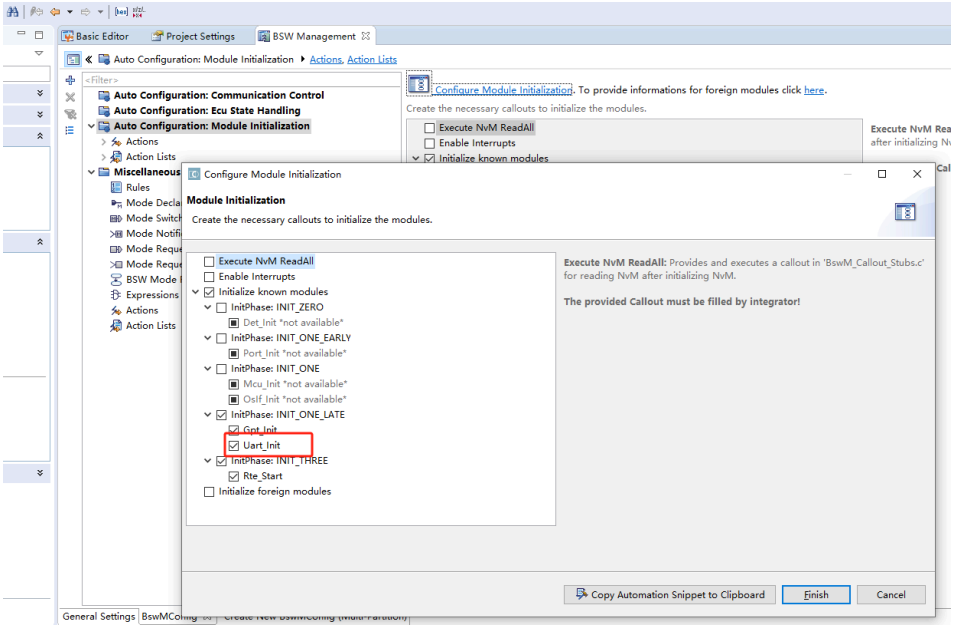
Address	Vector (all cores)	IRQ	NVIC IPR register number	NVIC non-IPR register number	INTM support	Instance	Interrupt request
							Receive Interrupt
0x0000028C	163	147	36	4	Y	LPUART_6	Transmit Interrupt
							Receive Interrupt
0x00000290	164	148	37	4	Y	LPUART_7	Transmit Interrupt
							Receive Interrupt
0x00000294	165	149	37	4	Y	LPUART_8	Transmit Interrupt
							Receive Interrupt
0x00000298	166	150	37	4	Y	LPUART_9	Transmit Interrupt
							Receive Interrupt
0x0000029C	167	151	37	4	Y	LPUART_10	Transmit Interrupt
							Receive Interrupt
0x000002A0	168	152	38	4	Y	LPUART_11	Transmit Interrupt
							Receive Interrupt
0x000002A4	169	153	38	4	Y	LPUART_12	Transmit Interrupt
							Receive Interrupt
0x000002A8	170	154	38	4	Y	LPUART_13	Transmit Interrupt
							Receive Interrupt
0x000002AC	171	155	38	4	Y	LPUART_14	Transmit Interrupt
							Receive Interrupt
0x000002B0	172	156	39	4	Y	LPUART_15	Transmit Interrupt
							Receive Interrupt
0x000002B4	173	157	39	4		RESERVED	
0x000002B8	174	158	39	4		RESERVED	
0x000002BC	175	159	39	4		RESERVED	
0x000002C0	176	160	40	5	Y	I3C_0	I3C interrupt
0x000002C4	177	161	40	5	Y	LPI2C_0	LPI2C Master Interrupt
							LPI2C Interrupt
0x000002C8	178	162	40	5	Y	LPI2C_1	LPI2C Master Interrupt
							LPI2C Interrupt
0x000002CC	179	163	40	5		RESERVED	
0x000002D0	180	164	41	5		RESERVED	
0x000002D4	181	165	41	5	Y	LPSPI_0	LPSPI Interrupt
0x000002D8	182	166	41	5	Y	LPSPI_1	LPSPI Interrupt
0x000002DC	183	167	41	5	Y	LPSPI_2	LPSPI Interrupt
0x000002E0	184	168	42	5	Y	LPSPI_3	LPSPI Interrupt
0x000002E4	185	169	42	5	Y	LPSPI_4	LPSPI Interrupt
0x000002E8	186	170	42	5	Y	LPSPI_5	LPSPI Interrupt
0x000002EC	187	171	42	5		RESERVED	
0x000002F0	188	172	43	5		RESERVED	
0x000002F4	189	173	43	5	Y	QSPI	TX Buffer Fill interrupt
							Transfer Complete / Transaction Finished
							RX Buffer Drain interrupt

中断号在数据手册附件的S32K3xx_interrupt_map.xlsx中查的为163

中断名在MCAL代码中找到



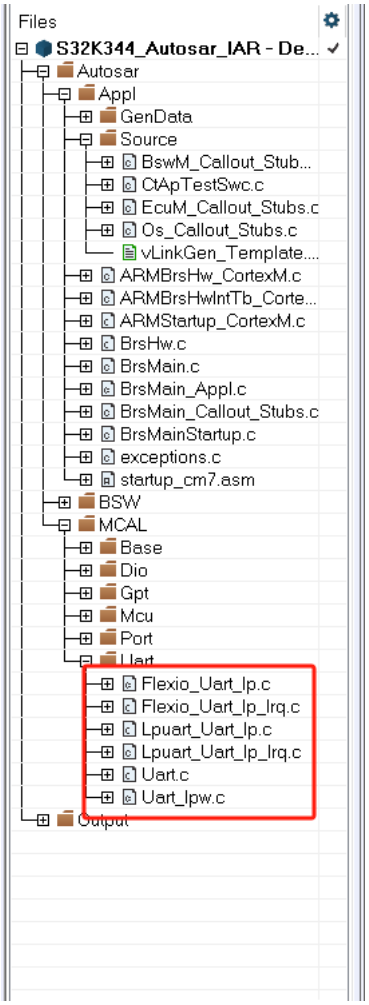
六、BswM添加uart_init



然后生成uart/port/os/rte/bswm

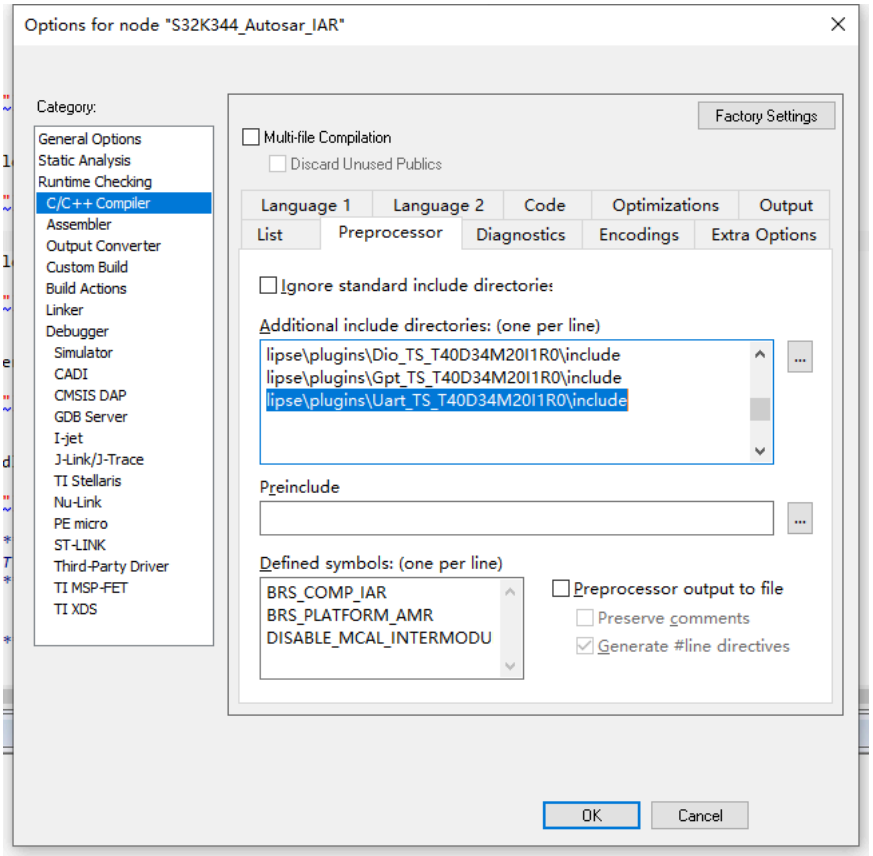
七、添加代码到IAR

添加MCAL的代码，新建Uart Group，然后添加mcal的c

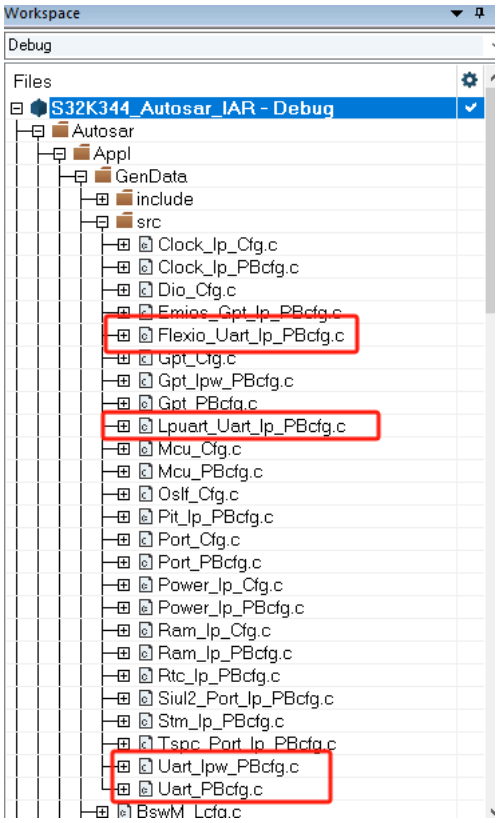


头文件路径加的IAR

$PROJ_DIR$\CBD2300384_D00\ThirdParty\Mcal_S32k\Supply\SW32K3_RTD_4.4_2.0.1\eclipse\plugins\Uart_TS_T40D34

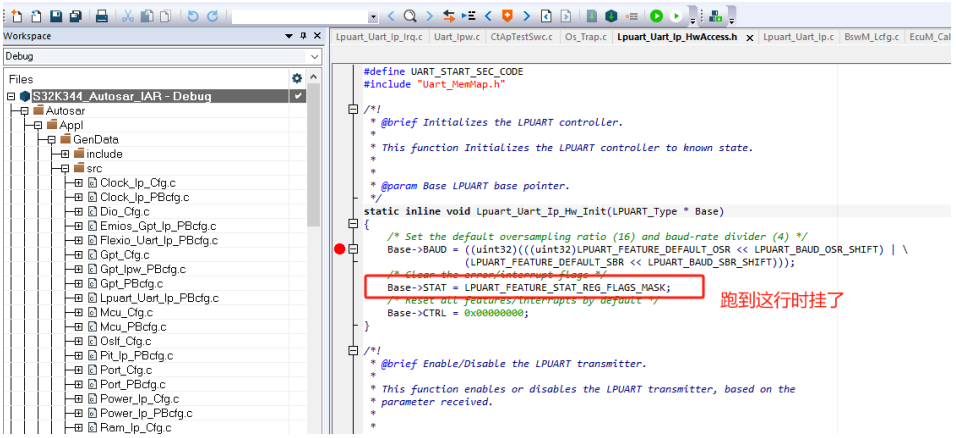


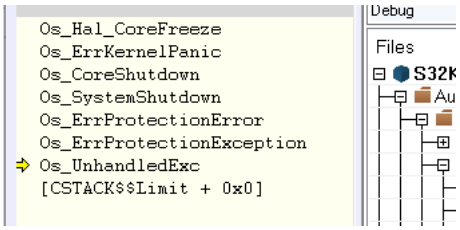
添加autosar生成的代码到 IAR



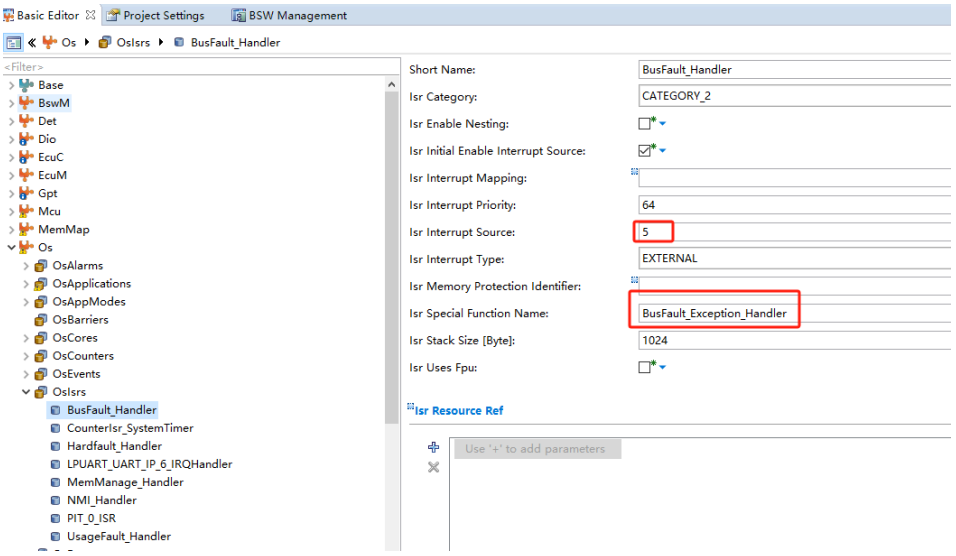
八、跑起来有问题！

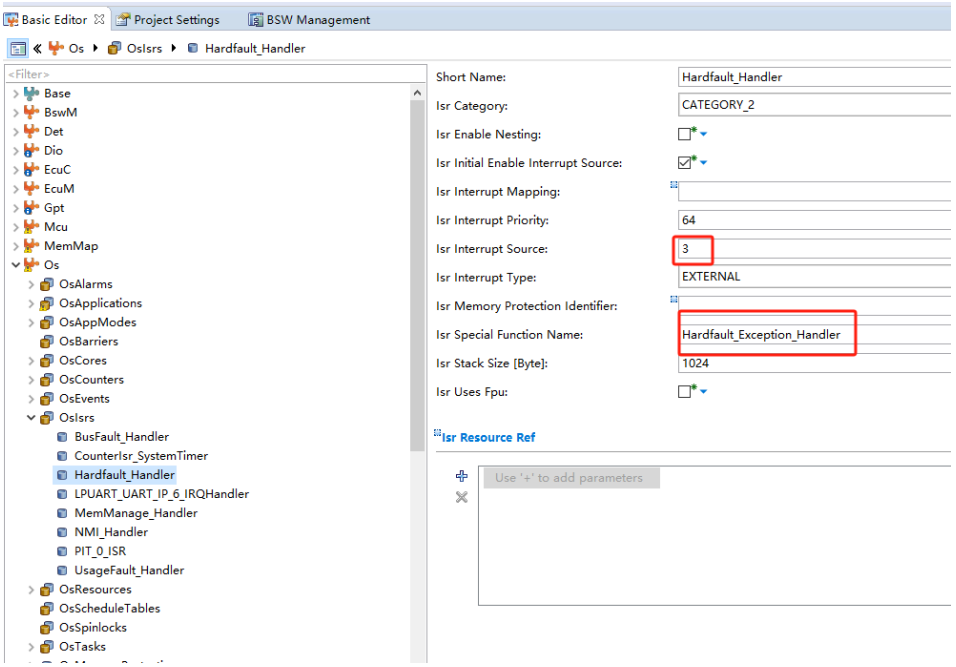
.

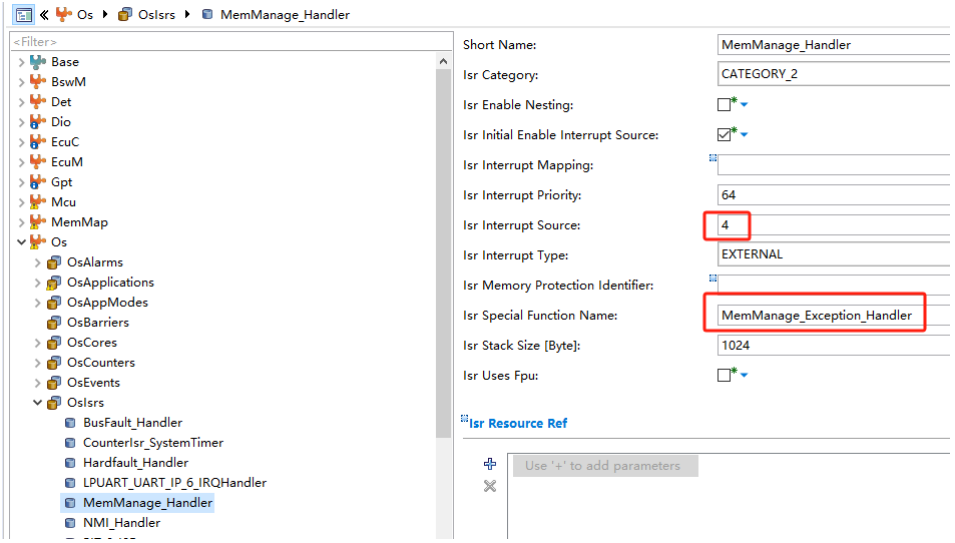


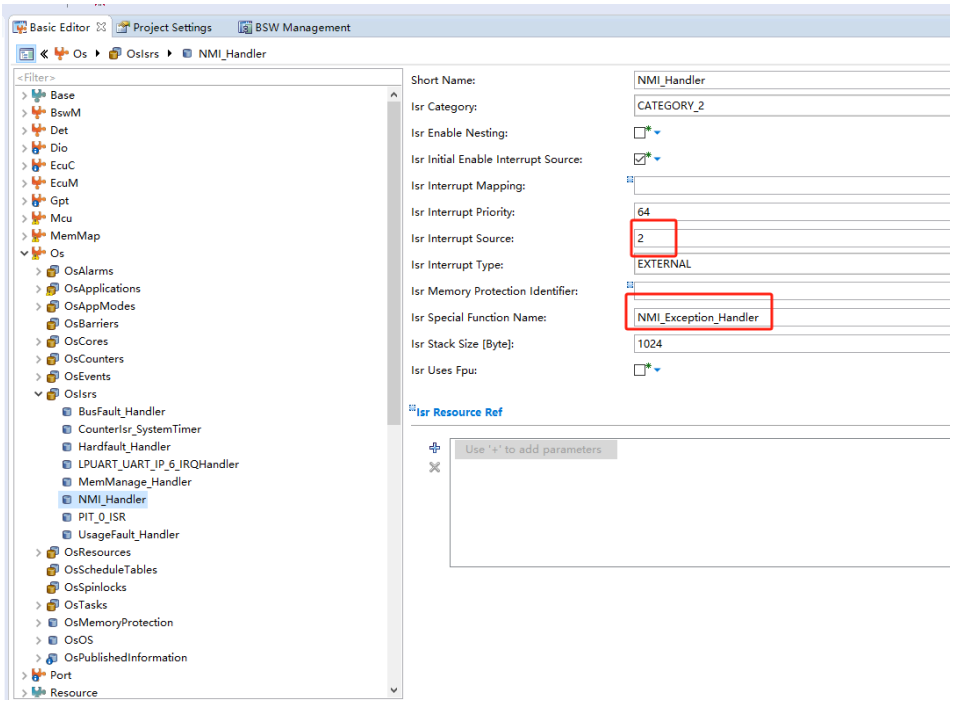


九、添加busfault\hardfault\memmange\NMI错误中断查看继续分析

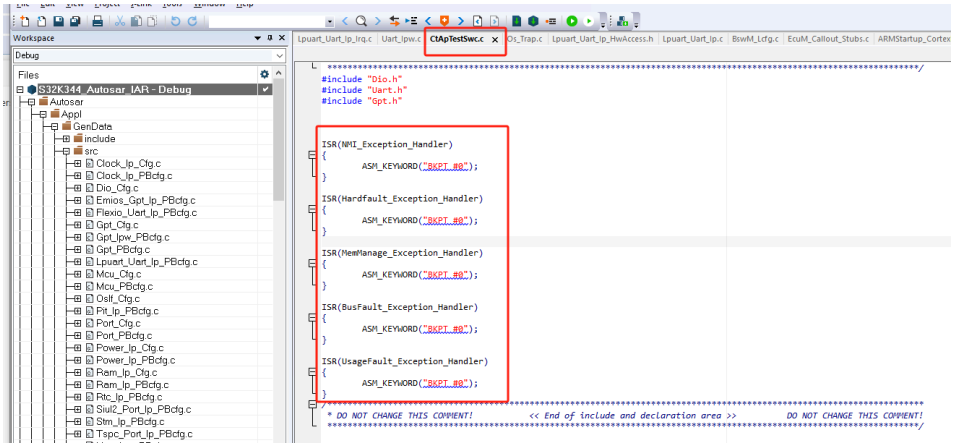



Basic Editor
Project Settings
BSW Management
Os
OsIsrs
Hardfault_Handler
<Filter>
Base
BswM
Det
Dio
EcuC
EcuM
Gpt
Mcu
MemMap
Os
OsAlarms
OsApplications
OsAppModes
OsBarriers
OsCores
OsCounters
OsEvents
OsIsrs
BusFault_Handler
CounterIsr_SystemTimer
Hardfault_Handler
LPUART_UART_IP_6_IRQHandler
MemManage_Handler
NMI_Handler
PIT_0_ISR
UsageFault_Handler
OsResources
OsScheduleTables
OsSpinlocks
OsTasks
Short Name:
Hardfault_Handler
Isr Category:
CATEGORY_2
Isr Enable Nesting:
Isr Initial Enable Interrupt Source:
Isr Interrupt Mapping:
Isr Interrupt Priority:
64
Isr Interrupt Source:
3
Isr Interrupt Type:
EXTERNAL
Isr Memory Protection Identifier:
Isr Special Function Name:
Hardfault_Exception_Handler
Isr Stack Size [Byte]:
1024
Isr Uses Fpu:
Isr Resource Ref
Use '+' to add parameters

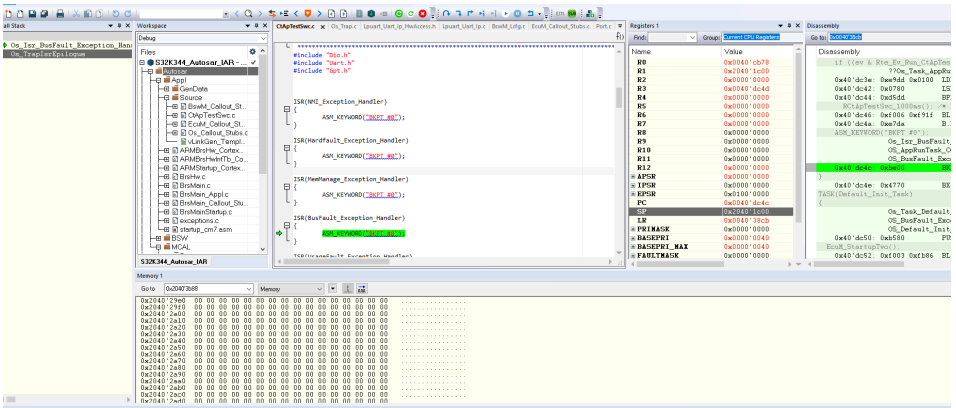

Os
OsIsrs
MemManage_Handler
<Filter>
Base
BswM
Det
Dio
EcuC
EcuM
Gpt
Mcu
MemMap
Os
OsAlarms
OsApplications
OsAppModes
OsBarriers
OsCores
OsCounters
OsEvents
OsIsrs
BusFault_Handler
CounterIsr_SystemTimer
Hardfault_Handler
LPUART_UART_IP_6_IRQHandler
MemManage_Handler
NMI_Handler
Short Name:
MemManage_Handler
Isr Category:
CATEGORY_2
Isr Enable Nesting:
Isr Initial Enable Interrupt Source:
Isr Interrupt Mapping:
Isr Interrupt Priority:
64
Isr Interrupt Source:
4
Isr Interrupt Type:
EXTERNAL
Isr Memory Protection Identifier:
Isr Special Function Name:
MemManage_Exception_Handler
Isr Stack Size [Byte]:
1024
Isr Uses Fpu:
Isr Resource Ref
Use '+' to add parameters



添加对应的中断函数，先在CtApTestSwc.c中添加

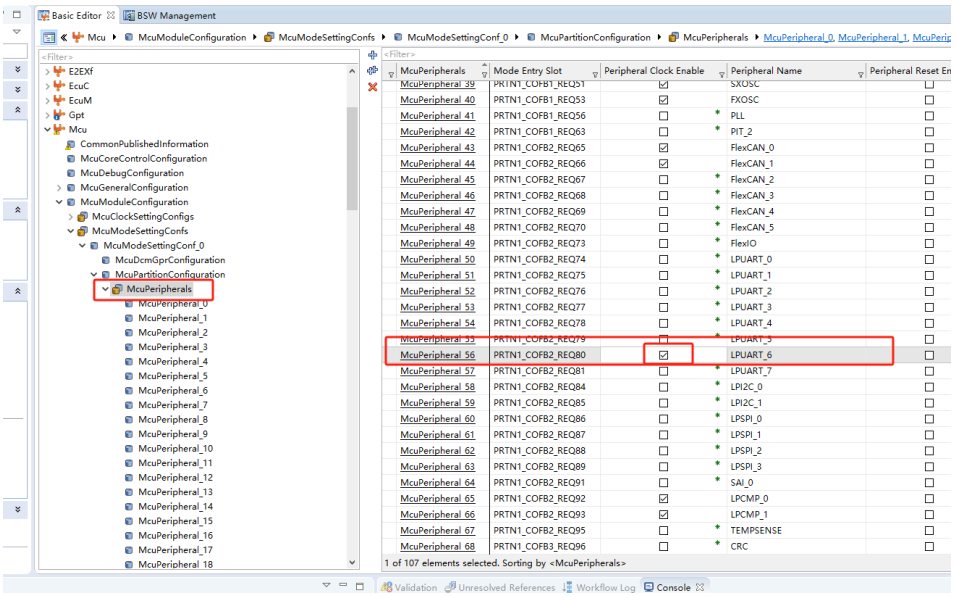


最后跑死在BusFault, 待后续分析



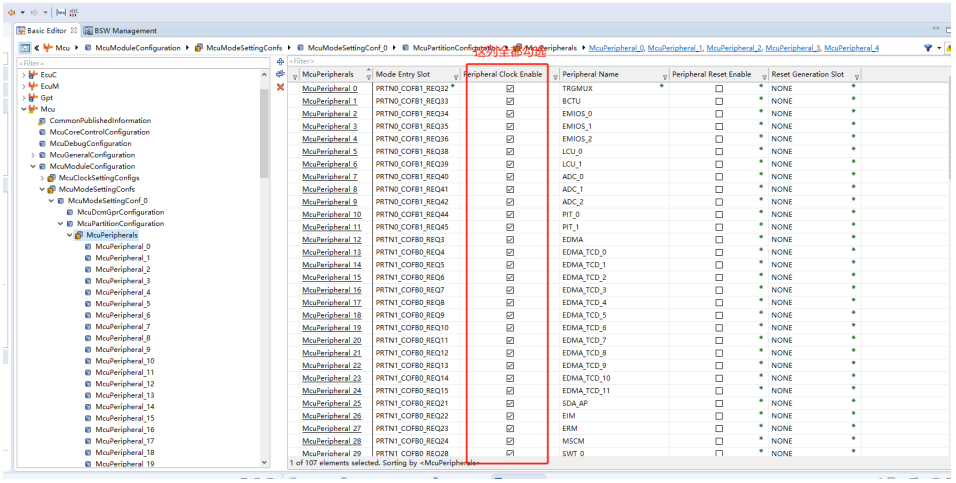
十、后续分析解决BusFault问题

最后定位是没有把串口6的外设时钟打开导致的，需要在mcu模块中把串口6的外设时钟打开即可



重新生成MCU模块即可正常跑起来

注：为了后续不因为外设时钟没开而影响调试，现在这里把所有外设时钟都勾上



十一、使用printf 串口打印

随便找一个.c 里面添加三个头文件，并且重写printf的实现即可，里面调用到Uart_SyncSend 串口打印函数

```
BswM.c | Os_Trap.c | Can.c | CtApTestSwc.c * | EcuM_Callout_Stubs.c | BswM_Lcfg.c | ARMStartup_CortexM.c

/*****************************************************************************
 * DO NOT CHANGE THIS COMMENT!           << End of version logging area >>
 *****************************************************************************

#include "Rte_CtApTestSwc.h"


/*****************************************************************************
 * DO NOT CHANGE THIS COMMENT!           << Start of include and declaration area
 *****************************************************************************
#include "Dio.h"
#include "Uart.h"
#include "Gpt.h"
#include <stdio.h>
#include <string.h>
#include <stdarg.h>
char u8PrintfBuf[1024] = {0};
int printf(const char *format,...)
{
    va_list args;
    va_start(args, format);
    vsprintf(u8PrintfBuf, format, args);
    va_end(args);
    Uart_SyncSend(0, u8PrintfBuf,strlen(u8PrintfBuf),0xFFFF);
    return 0;
}
```

然后在其他文件中，只要#include <stdio.h> , 然后正常调用printf 即可打印到串口